CCNA QUICK REVISION CHARTS

Networking basics • Security • Ethernet • Fiber • Cabling

1. Security concepts and firewall types

Term	Simple meaning	Remember
Host-based firewall	Firewall software on one PC/server.	Protects one endpoint.
Network-based firewall	Firewall placed in the network path/boundary.	Can protect many devices.
Packet-filtering firewall	Checks packet details such as IP addresses, protocol and ports.	Basic rule-based filtering.
Stateful firewall	Tracks active connections and their state.	Remembers the conversation.
Application / proxy firewall	Inspects application-layer traffic.	Layer 7 inspection.
NGFW	Next-Generation Firewall; may combine firewalling with application control, IPS and URL filtering.	Advanced, application-aware firewall.

হোস্টভিত্তিক ফায়ারওয়াল একটি কম্পিউটারকে রক্ষা করে। নেটওয়ার্কভিত্তিক ফায়ারওয়াল নেটওয়ার্কের ট্রাফিক নিয়ন্ত্রণ করে। স্টেটফুল ফায়ারওয়াল সংযোগের অবস্থা মনে রাখে।

Other security terms

Term	Meaning / job
Antivirus / Antispyware	Finds and helps remove malware or spyware.
IDS	Intrusion Detection System — detects suspicious activity and alerts.
IPS	Intrusion Prevention System — detects and can block suspicious activity.
VPN	Creates an encrypted tunnel over an untrusted network.
ACL	Access Control List — rules that permit or deny traffic.
AAA	Authentication (who are you?), Authorization (what may you do?), Accounting (what did you do?).
Encryption	Makes data unreadable to unauthorised parties.

Core network roles

Term	Definition	Example
Client	Requests a service; a role, not a specific device.	Browser requesting a website.
Server	Provides a service to clients.	Web server.
Node	Any device connected to a network.	PC, printer, router, switch.

2. Ethernet speeds and IEEE standards

Ethernet type	Speed	Mbps equivalent	IEEE standard	Medium
Ethernet	10 Mbps	10	802.3	Copper
Fast Ethernet	100 Mbps	100	802.3u	Copper
Gigabit (1000BASE-T)	1 Gbps	1,000	802.3ab	Copper
Gigabit (1000BASE-SX/LX)	1 Gbps	1,000	802.3z	Fiber
10 Gig (10GBASE-T)	10 Gbps	10,000	802.3an	Copper
10 Gig (10GBASE-SR/LR/ER)	10 Gbps	10,000	802.3ae	Fiber

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Bits, bytes and data rates

Unit	Value
1 kilobit (Kb)	1,000 bits
1 megabit (Mb)	1,000,000 bits
1 gigabit (Gb)	1,000,000,000 bits
1 byte (B)	8 bits
100 Mbps	100 ÷ 8 = 12.5 MB/s (theoretical, before overhead)

Notation	Means	Used for
b (small b)	bit	Network speed, e.g. Mbps
B (capital B)	byte	File size / transfer rate, e.g. MB

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Standards to memorise

Standard	Association
802.3u	100 Mbps Fast Ethernet
802.3ab	1 Gbps copper (1000BASE-T)
802.3z	1 Gbps fiber (1000BASE-SX/LX)
802.3an	10 Gbps copper (10GBASE-T)
802.3ae	10 Gbps fiber (10GBASE-SR/LR/ER)

Note: 10GBASE-ZR is commonly a vendor-specific long-reach optic; it is not one of the original 802.3ae SR/LR/ER variants.

3. Fiber-optic cables and reach

Feature	Single-mode fiber (SMF)	Multimode fiber (MMF)
Core size (typical)	About 9 µm	50 µm or 62.5 µm
Light path	One propagation mode	Multiple propagation modes
Distance	Long distance	Shorter distance
Light source	Laser	VCSEL / LED, depending on optic
Common use	Campus, metro and long-distance links	Short links inside buildings / data centres

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1-Gigabit fiber standards

Standard	Fiber type	Typical reach
1000BASE-SX	Multimode	Up to 550 m (depends on fiber grade)
1000BASE-LX	Multimode	Up to 550 m (suitable fiber / installation)
1000BASE-LX	Single-mode	Up to 5 km

10-Gigabit fiber reach

Standard	Meaning	Fiber	Typical reach
10GBASE-SR	Short Reach	Multimode	300 m on OM3; 400 m on OM4
10GBASE-LR	Long Reach	Single-mode	10 km
10GBASE-ER	Extended Reach	Single-mode	30 km
10GBASE-ZR	Very long reach	Single-mode	About 80 km; commonly vendor-specific

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Transceiver modules

Term	Quick meaning
SFP	Small Form-Factor Pluggable — removable transceiver module used in a switch/router port.
GBIC	An older, physically larger pluggable transceiver format.

Fiber connects to the optical transceiver; the transceiver plugs into the switch/router.

4. UTP cabling, pin pairs and cable selection

Ethernet type	Pairs used	Pin pairs
10BASE-T / 100BASE-TX	2 pairs	1–2 and 3–6
1000BASE-T	4 pairs	1–2, 3–6, 4–5 and 7–8
10GBASE-T	4 pairs	1–2, 3–6, 4–5 and 7–8

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Classic 10/100 Mbps TX/RX pin mapping

Port type	Transmit (TX)	Receive (RX)	Typical device
MDI	Pins 1, 2	Pins 3, 6	PC / router
MDI-X	Pins 3, 6	Pins 1, 2	Switch / hub

This TX/RX mapping is for classic 10/100 Mbps MDI/MDI-X connections. 1000BASE-T uses all four pairs bidirectionally.

Straight-through vs crossover

Cable	Wiring	Classic examples
Straight-through	Same pin number connects to the same pin number.	PC ↔ Switch; Router ↔ Switch (MDI ↔ MDI-X)
Crossover	Transmit/receive pairs are crossed (1–2 ↔ 3–6).	PC ↔ PC; Switch ↔ Switch; same port role

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Cable properties

Cable	EMI protection	Key point
UTP	Lower; no shielding	Twisting reduces noise, but it is more vulnerable to EMI.
STP	Better shielding	Metal shield adds protection against external interference.
Fiber-optic	Immune to EMI	Carries light, not electrical signals.

Fact	Remember
Maximum Ethernet UTP channel length	100 m (commonly 90 m permanent link + up to 10 m patch cords)
Why UTP pairs are twisted	To reduce EMI and crosstalk.
Auto MDI-X	Automatically adjusts transmit/receive pair mapping.

LAST-MINUTE MEMORY SHEET

A fast scan before a CCNA quiz or practice test

Question clue	Answer to recall
100 Mbps standard	802.3u
1 Gbps copper / fiber	802.3ab / 802.3z
10 Gbps copper / fiber	802.3an / 802.3ae
1 Byte	8 bits
UTP maximum Ethernet channel length	100 m
10GBASE-SR / LR / ER	300 m on OM3 / 10 km / 30 km
Two fiber types	Single-mode and multimode
Crossover cable	Crosses transmit/receive pairs
Auto MDI-X	Automatically adjusts pin-pair mapping
1000BASE-T / 10GBASE-T	Uses all four copper pairs
SFP	Small pluggable transceiver module
UTP / STP / Fiber	Unshielded / shielded / immune to EMI
Stateful firewall	Tracks active connection state
IDS / IPS	Detects / detects and can prevent
AAA	Authentication / Authorization / Accounting

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Accuracy notes: 10GBASE-SR reach depends on fiber grade (OM3: 300 m; OM4: 400 m). 10GBASE-ZR is commonly vendor-specific. Classic TX/RX pin mapping is for 10/100 Mbps MDI/MDI-X ports.